

Fitting Lines and Curves

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Topic: Boundary Detection, Module: Features

First Principles of Computer Vision

Preprocessing Edge Images



Edge
Detection



Thresholding



Shrink
& Expand



Manually Sketched



Boundary
Detection



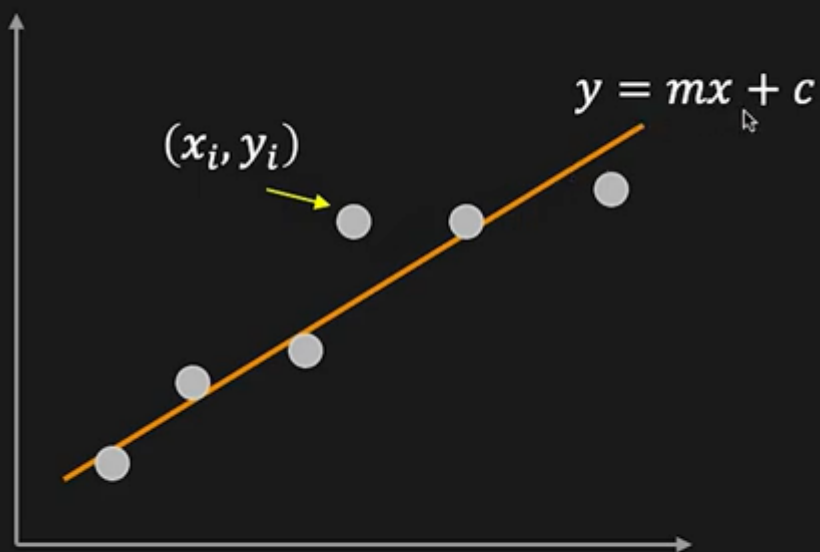
Thinning



Fitting Lines to Edges

Given: Edge Points (x_i, y_i)

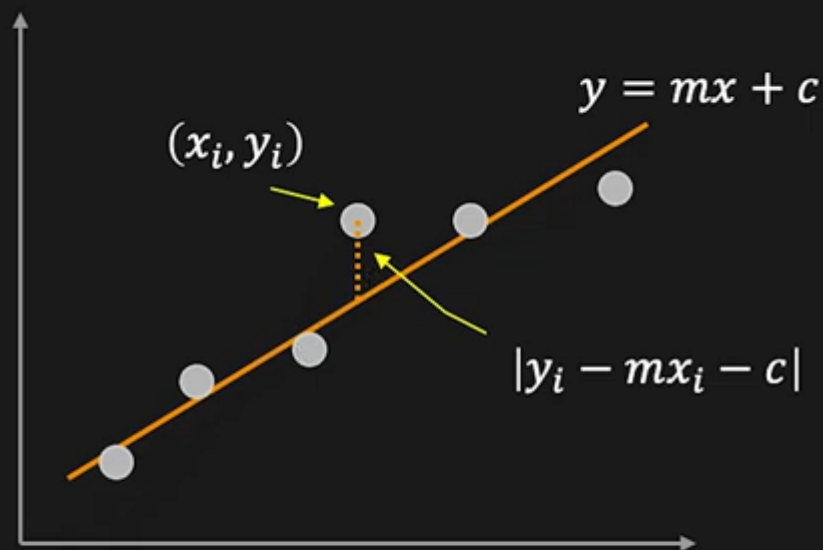
Task: Find (m, c)



Fitting Lines to Edges

Given: Edge Points (x_i, y_i)

Task: Find (m, c)



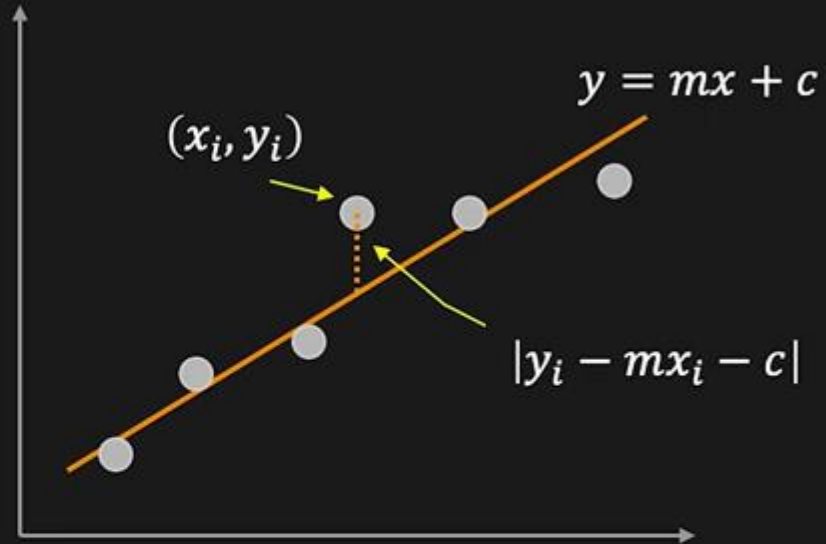
Minimize: Average Squared **Vertical** Distance

$$E = \frac{1}{N} \sum_i (y_i - mx_i - c)^2$$

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Least Squares Solution:

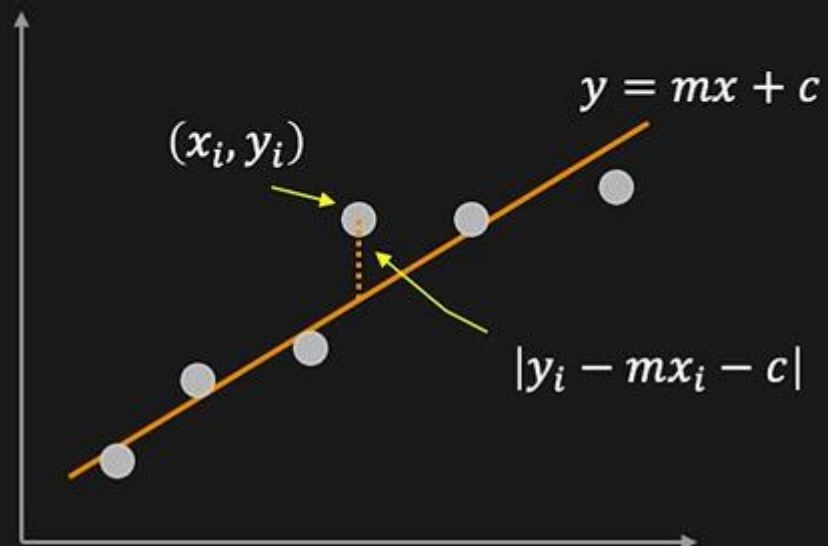
$$\frac{\partial E}{\partial m} = \frac{-2}{N} \sum_i x_i (y_i - mx_i - c) = 0$$

$$\frac{\partial E}{\partial c} = \frac{-2}{N} \sum_i (y_i - mx_i - c) = 0$$

Fitting Lines to Edges

Given: Edge Points (x_i, y_i)

Task: Find (m, c)



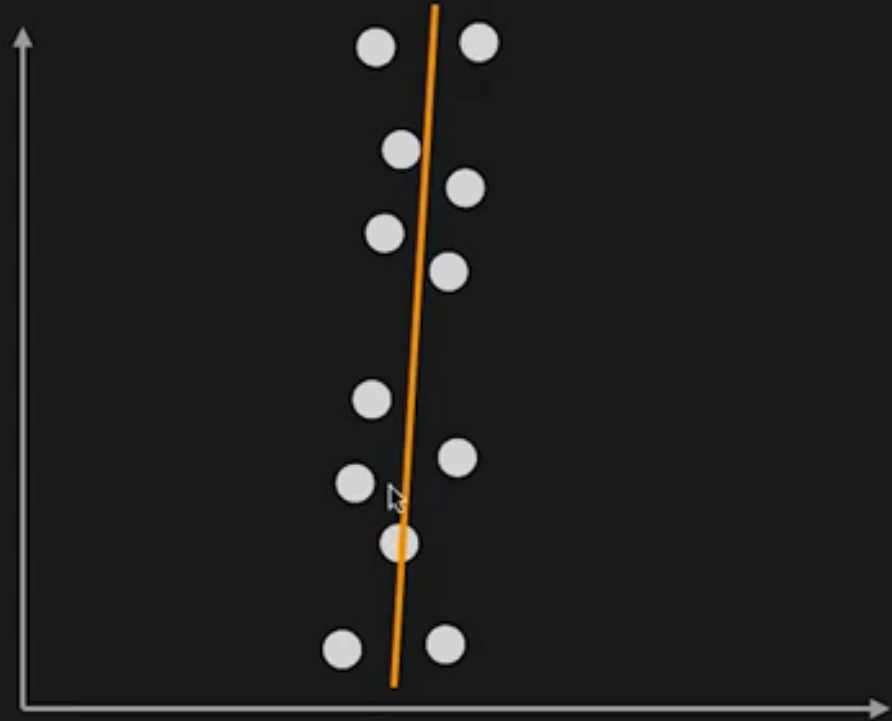
Solution:

$$m = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} \quad c = \bar{y} - m\bar{x}$$

$$\text{where: } \bar{x} = \frac{1}{N} \sum_i x_i \quad \bar{y} = \frac{1}{N} \sum_i y_i$$

Fitting Lines to Edges

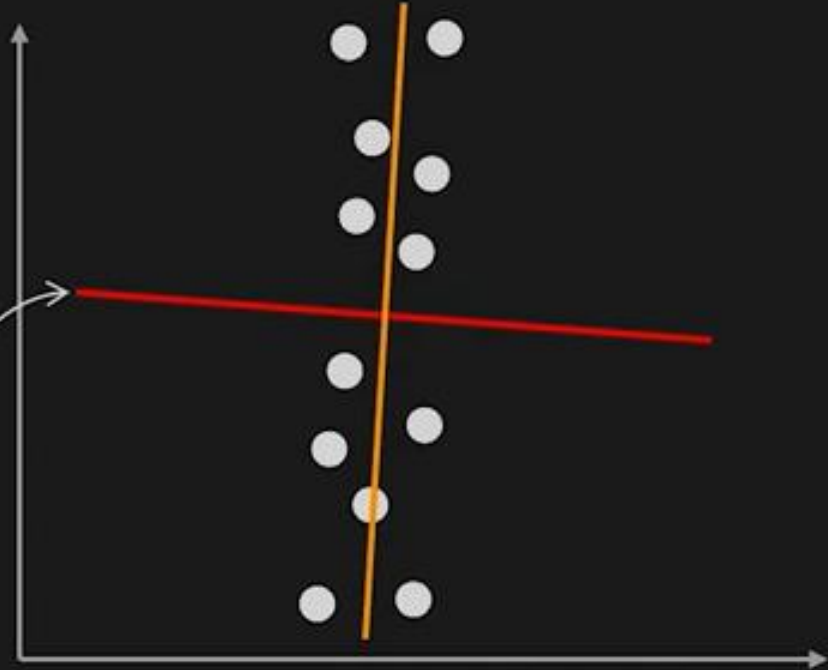
Problem: When the points represent a vertical line.



Fitting Lines to Edges

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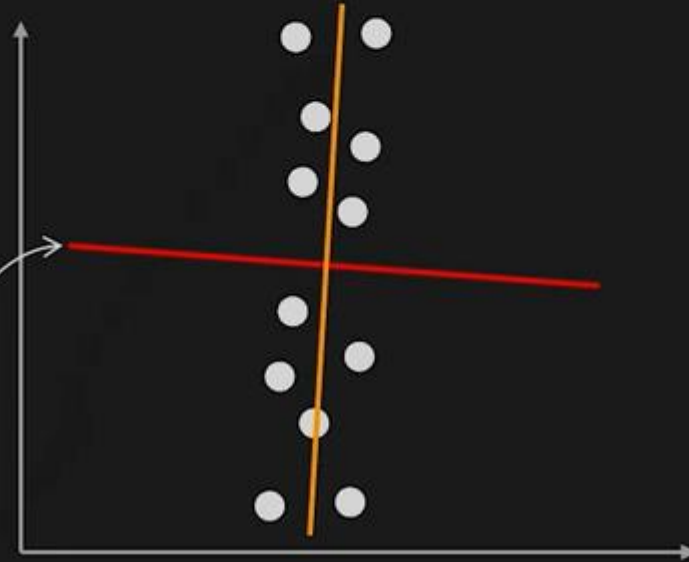
Line that minimizes E !



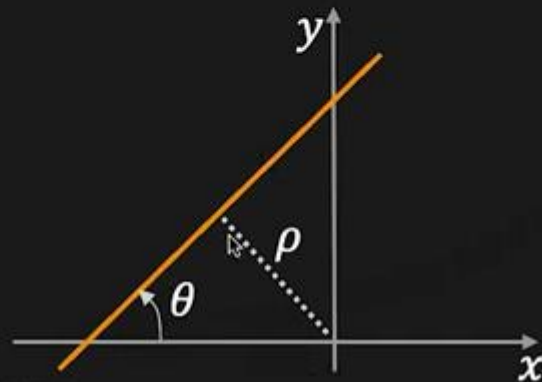
Fitting Lines to Edges

Problem: When the points represent a vertical line.

Line that minimizes E!



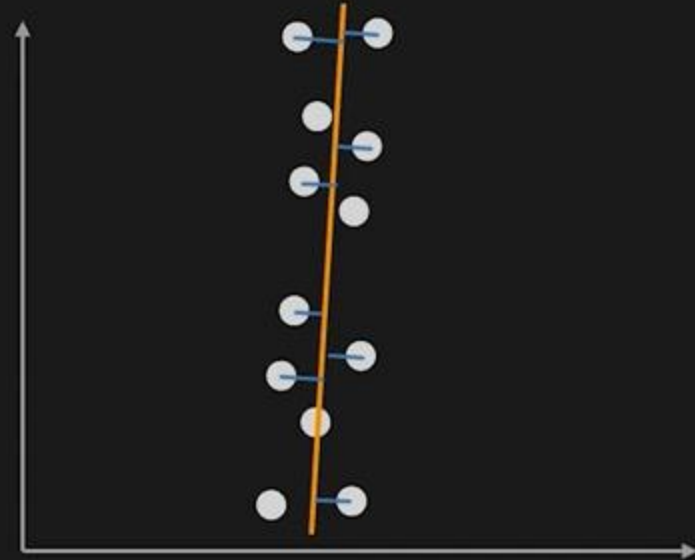
Solution: Use a different line equation



$$x \sin \theta - y \cos \theta + \rho = 0$$

Fitting Lines to Edges

Problem: When the points represent a vertical line.



Minimize: Average Squared **Perpendicular** Distance

$$E = \frac{1}{N} \sum_i \underbrace{(x_i \sin \theta - y_i \cos \theta + \rho)^2}_{\text{Perpendicular Distance}^2}$$

(See Binary Img. Processing Lecture)

Fitting Curves to Edges

Given: Edge Points (x_i, y_i)

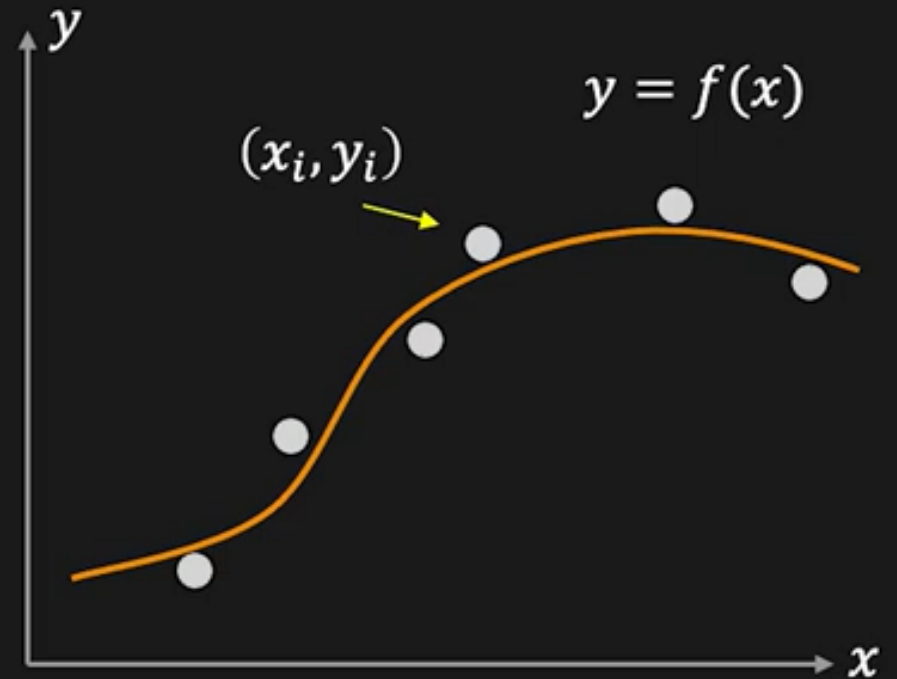
Task: Find polynomial

$$y = f(x) = ax^3 + bx^2 + cx + d$$

that best fits the points

Minimize:

$$E = \frac{1}{N} \sum_i (y_i - ax_i^3 - bx_i^2 - cx_i - d)^2$$



Fitting Curves to Edges

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Task: Find polynomial

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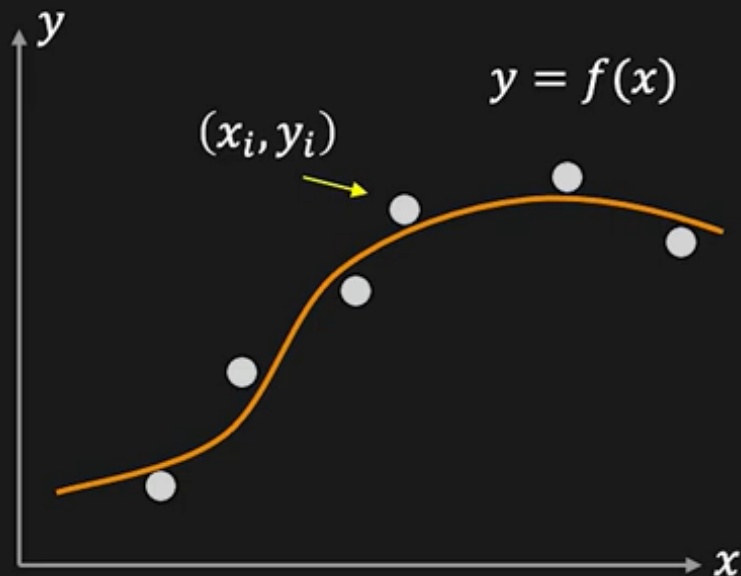
Minimize:

$$E = \frac{1}{N} \sum_i (y_i - ax_i^3 - bx_i^2 - cx_i - d)^2$$

Solve the Linear System Using Least Squares Fit by:

$$\frac{\partial E}{\partial a} = 0 \quad \frac{\partial E}{\partial b} = 0 \quad \frac{\partial E}{\partial c} = 0 \quad \frac{\partial E}{\partial d} = 0$$

Closed-form solution cumbersome when unknowns are many



Fitting Curves to Edges

Solving as a Linear System:

$$y_0 = ax_0^3 + bx_0^2 + cx_0 + d$$

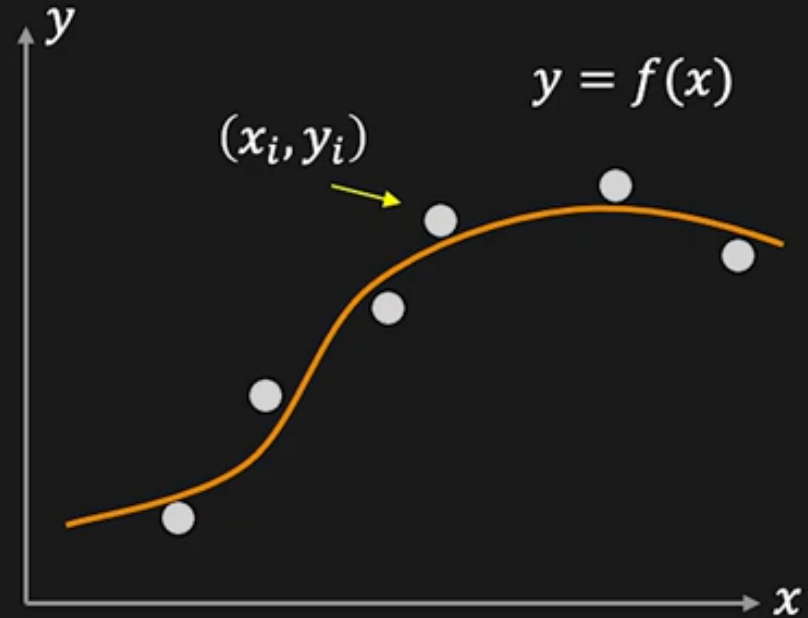
$$y_1 = ax_1^3 + bx_1^2 + cx_1 + d$$

$$\vdots$$

$$y_i = ax_i^3 + bx_i^2 + cx_i + d$$

$$\vdots$$

$$y_n = ax_n^3 + bx_n^2 + cx_n + d$$



Given many (x_i, y_i) 's, this is an over-determined linear system with four unknowns (a, b, c, d) .

Solving a Linear System

An over-determined linear system with m unknowns $\{a_j\}$ ($j = 0, \dots, m$) and n observations $\{(x_{ij}, y_i)\}$ ($i = 0, \dots, n$) ($n > m$) can be written in a matrix form.

$$\begin{bmatrix} x_{00} & x_{01} & \dots & x_{0m} \\ x_{10} & x_{11} & \dots & x_{1m} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n0} & x_{n1} & \dots & x_{nm} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_m \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Solving a Linear System

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$X_{n \times m}$ Known $\mathbf{a}_{m \times 1}$ Unknown $\mathbf{y}_{n \times 1}$ Known

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Least Squares Solution:

$$X^T X \mathbf{a} = X^T \mathbf{y} \quad \Rightarrow \quad \mathbf{a} = (X^T X)^{-1} X^T \mathbf{y} \quad X^+ = (X^T X)^{-1} X^T$$

$\mathbf{a} = X^+ \mathbf{y}$ (Pseudo Inverse)

Reference

- <https://www.youtube.com/watch?v=QQI4jUYRUc8>